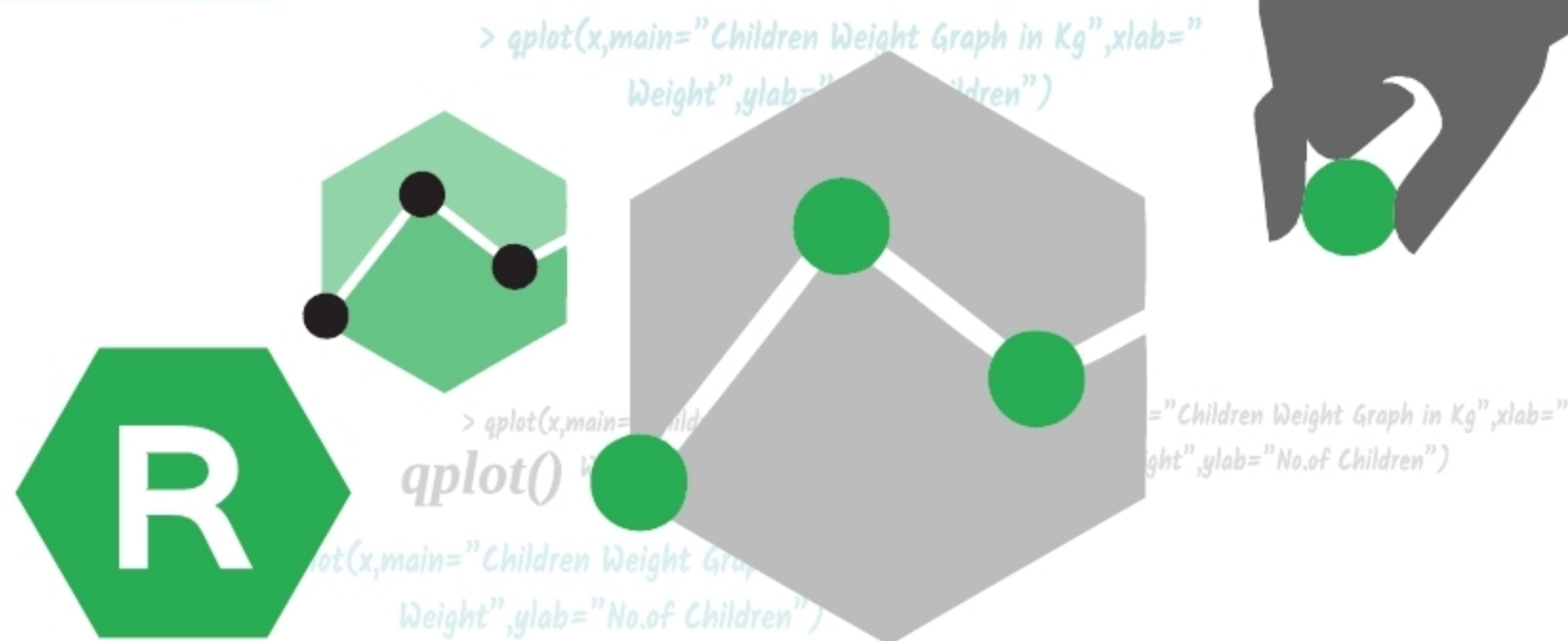




Plot Quickly and Efficiently with **ggplot2** in R

R is an excellent environment for visualisation. The well-designed ggplot2 package with a planned structure for any kind of graphics is available in R. The *qplot()* function of the ggplot2 bundle can play a key role in producing fast and proficient plots.

R programming offers an exceptional set of libraries to build astonishing visualisations to represent data. The most accepted and valuable libraries of visualisation in R are ggplot2, lattice, higher charter, leaflet, RGL and Ploty. Of these, ggplot2 is the trendiest package, which was written by Hadley Wickham (<http://hadley.nz/>). The ggplot2 package has a vital implementation of the grammar of graphics. The basic elements of the grammar of ggplot2 are depicted in Table 1.

Table 1: Basic elements of the grammar of graphics in ggplot2

Basic element	Description
Data	This illustrates the data that you want to plot in the graph.
Geometries	This represents the visual elements that we need for the data.
Aesthetics	This covers the colour, size, shape and position onto which we plot our data. Generally, it is applicable on statistical and geometric objects.
Facet	This is the cluster by which we split the data.

In addition to the basic elements of the grammar of graphics, ggplot2 also contains some advanced elements of the grammar of graphics. The ggplot2 package has the

capability to plot simple as well as complex graphs based on the problem statement. This article will focus on simple and quick graph generation using ggplot2. With the help of the function *qplot()* of ggplot2, we can create a variety of plots effortlessly. We will then discuss the installation and basic structure of the ggplot2 library before moving on to the *qplot()* function and, eventually, look at useful options and how to generate different categories of graphs using *qplot()* with examples.

Installation and basic structure of ggplot2

ggplot2 is not the component of the regular distribution of R, and hence it is to be installed explicitly by using the *install.packages()* command. To install ggplot2, the following syntax is to be used:

```
> install.packages("ggplot2")
```

When you enter the above command on the R console, the installation process will start and ask for the HTTP CRAN mirror. Select *India(https)* from the menu, as is shown in Figure 1.

Once it is installed, load the ggplot2 library with the following command:

```
> library("ggplot2")
```